

BML lecture #1: Bayesics

<http://github.com/rbardenet/bml-course>

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What comes to *your* mind when you hear "Bayesian ML"?

- 1** Introduction
- 2** A warmup: Estimation in regression models
- 3** ML as data-driven decision-making
- 4** Subjective expected utility
- 5** Specifying joint models
- 6** 50 shades of Bayes

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A quick motivating example before we go formal 1/2

- ▶ Let N individuals evolve from Susceptible to Infected to Recovered, $x_n(t) \in \{S, I, R\}$, $1 \leq n \leq N$, $t \in [0, T]$.
- ▶ Each susceptible individual n moves to I according to a Poisson process with intensity

$$\sum_{k: x_k(t)=I} \lambda_{nk}(\theta_{SI}).$$

- ▶ Each infected person recovers after a Gamma(a, b) time.
- ▶ This allows to express

$$p(x_1(t_{1,1}), \dots, x_1(t_{1,T_1}), \dots, x_N(t_{N,1}), \dots, x_1(t_{N,T_N}) | \theta).$$

where $\theta = (\theta_{SI}, a, b)$.

- ▶ Now, consider $p(\theta | \text{data}) \propto p(\text{data} | \theta) p(\theta)$.

A quick motivating example before we go formal 2/2

- ▶ If asked to report an interval A on a particular function of θ , say $R_0 = h(\theta)$, I would report a small interval A such that

$$\int 1_{h(\theta) \in A} p(\theta | \text{data}) d\theta = p(h(\theta) \in A | \text{data}) \geq 0.95.$$

- ▶ If asked whether we should close universities, I would ask for
 - ▶ the cost α of closing unis when $R_0 < 1$,
 - ▶ the cost β of keeping unis open while $R_0 > 1$.
- ▶ Then I would recommend closing if and only if

$$p(R_0 > 1 | \text{data}) > \frac{\alpha}{\alpha + \beta}.$$

- ▶ Additionally, I would check that the decision doesn't change if I change my prior $p(\theta)$ a little.
- ▶ If it did, then I would refine my likelihood and/or wait for more data.

- ▶ *[...] practical methods for making inferences from data, using probability models for quantities we observe and for quantities about which we wish to learn.*
- ▶ *The essential characteristic of Bayesian methods is their explicit use of probability for quantifying uncertainty in inferences based on statistical data analysis.*
- ▶ *Three steps:*
 - 1 *Setting up a full probability model,*
 - 2 *Conditioning on observed data, calculating and interpreting the appropriate “posterior distribution”,*
 - 3 *Evaluating the fit of the model and the implications of the resulting posterior distribution. In response, one can alter or expand the model and repeat the three steps.*

- ▶ $y_{1:n} = (y_1, \dots, y_n) \in \mathcal{Y}^n$ denote observable data/labels.
- ▶ $x_{1:n} \in \mathcal{X}^n$ denote covariates/features/hidden states.
- ▶ $z_{1:n} \in \mathcal{Z}^n$ denote hidden variables.
- ▶ $\theta \in \Theta$ denote parameters.
- ▶ X denotes an $\mathcal{X}$ -valued random variable. Lowercase x denotes either a point in $\mathcal{X}$ or an $\mathcal{X}$ -valued random variable.

- ▶ Whenever it can easily be made formal, we write densities for our random variables and let the context indicate what is meant. So if $X \sim \mathcal{N}(0, \sigma^2)$, we write

$$\mathbb{E}h(X) = \int h(x) \frac{e^{-x^2/2\sigma^2}}{\sigma\sqrt{2\pi}} dx = \int h(x)p(x)dx.$$

Similarly, for $X \sim \mathcal{P}(\lambda)$, we write

$$\mathbb{E}h(X) = \sum_{k=0}^{\infty} h(k) e^{-\lambda} \frac{\lambda^k}{k!} = \int h(x)p(x)dx$$

- ▶ All pdfs are denoted by p , so that, e. g.

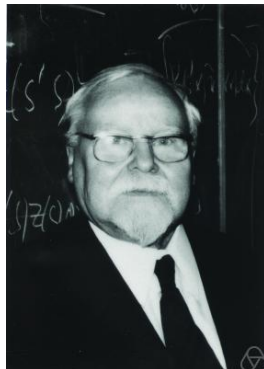
$$\begin{aligned}\mathbb{E}h(Y, \theta) &= \int h(y, \theta)p(y, \theta) dyd\theta \\ &= \int h(y, \theta)p(y, x, \theta) dx dy d\theta \\ &= \int h(y, \theta)p(y, \theta|x)p(x) dx dy d\theta\end{aligned}$$

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- ▶ A state space $\mathcal{S}$,
Every quantity you need to consider to make your decision.
- ▶ Actions $\mathcal{A} \subset \mathcal{F}(\mathcal{S}, \mathcal{Z})$,
Making a decision means picking one of the available actions.
- ▶ A reward space $\mathcal{Z}$,
Encodes how you feel about having picked a particular action.
- ▶ A loss function $L : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}_+$.
How much you would suffer from picking action a in state s .

- ▶ $\mathcal{S} = \mathcal{X}^n \times \mathcal{Y}^n \times \mathcal{X} \times \mathcal{Y}$, i.e. $s = (x_{1:n}, y_{1:n}, x, y)$.
- ▶ $\mathcal{Z} = \{0, 1\}$.
- ▶ $\mathcal{A} = \{a_g : s \mapsto 1_{y \neq g(x; x_{1:n}, y_{1:n})}, \quad g \in \mathcal{G}\}$.
- ▶ $L(a_g, s) = 1_{y \neq g(x; x_{1:n}, y_{1:n})}$.

PAC bounds; see e.g. (Shalev-Shwartz and Ben-David, 2014)

Let $(x_{1:n}, y_{1:n}) \sim \mathbb{P}^{\otimes n}$, and independently $(x, y) \sim \mathbb{P}$, we want an algorithm $g(\cdot; x_{1:n}, y_{1:n}) \in \mathcal{G}$ such that if $n \geq n(\delta, \varepsilon)$,

$$\mathbb{P}^{\otimes n} \left[\mathbb{E}_{(x,y) \sim \mathbb{P}} L(a_g, s) \leq \varepsilon \right] \geq 1 - \delta.$$

▶ $\mathcal{S} =$

▶ $\mathcal{Z} =$

▶ $\mathcal{A} =$

▶

► $\mathcal{S} =$

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The subjective expected utility principle

- 1 Choose $\mathcal{S}, \mathcal{Z}, \mathcal{A}$ and a loss function $L(a, s)$,
- 2 Choose a distribution p over $\mathcal{S}$,
- 3 Take the the corresponding Bayes action

$$a^* \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s \sim p} L(a, s). \quad (1)$$

Corollary: minimize the posterior expected loss

Now partition $s = (s_{\text{obs}}, s_{\text{u}})$, then

$$a^* \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s_{\text{obs}}} \mathbb{E}_{s_{\text{u}} | s_{\text{obs}}} L(a, s).$$

In ML, $\mathcal{A} = \{a_g\}$, with $g = g(s_{\text{obs}})$, so that (1) is equivalent to

$$a^* = \delta(s_{\text{obs}}) = \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s_{\text{u}} | s_{\text{obs}}} L(a, s).$$

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- ▶ PGMs (aka “Bayesian” networks) represent the dependencies in a joint distribution $p(y)$ by a directed graph $G = (E, V)$.
- ▶ Two important properties:

$$p(y) = \prod_{v \in V} p(y_v | y_{\text{pa}(v)}) \quad \text{and} \quad y_v \perp y_{\text{nd}(v)} | y_{\text{pa}(v)}.$$

A recap on probabilistic graphical models 2/2

- Also good to know how to determine whether $A \perp B | C$; see (Murphy, 2012, Section 10.5).

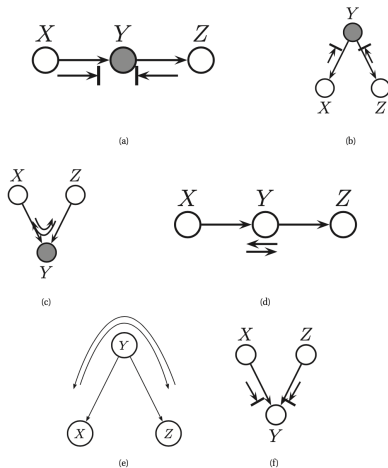


Figure 10.9 Bayes ball rules. A shaded node is one we condition on. If there is an arrow hitting a bar, it means the ball cannot pass through; otherwise the ball can pass through. Based on (Jordan 2007).

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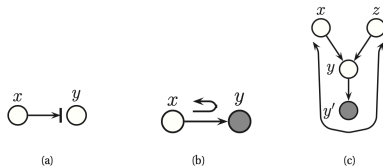


Figure 10.10 (a-b) Bayes ball boundary conditions. (c) Example of why we need boundary conditions. y' is an observed child of y , rendering y “effectively observed”, so the ball bounces back up on its way from x to z .

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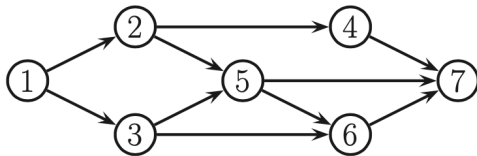


Figure 10.11 A DGM.

- ▶ Check that $x_2 \perp x_6 | x_5, x_1$, but that $x_2 \not\perp x_6 | x_5, x_1, x_7$.

Nonstandard Bayesian views challenge our prior/posterior interpretation

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An issue (or is it?)

Depending on how they interpret and how they implement SEU, you will meet many types of Bayesians (46656, according to Good).

A few divisive questions

- ▶ Using data or the likelihood to choose your prior; see Lecture #5.
- ▶ Using MAP estimators for their computational tractability, like in inverse problems

$$\hat{x}_\lambda \in \arg \min \|y - Ax\| + \lambda \Omega(x).$$

- ▶ When and how should you revise your model (likelihood or prior)?
- ▶ MCMC vs variational Bayes (more in Lectures #2 and #3)

- [1] A. Gelman et al. *Bayesian data analysis*. 3rd. CRC press, 2013.
- [2] K. Murphy. *Machine learning: a probabilistic perspective*. MIT Press, 2012.
- [3] S. Shalev-Shwartz and S. Ben-David. *Understanding machine learning: From theory to algorithms*. Cambridge university press, 2014.